

1. The Elbow Method

- We plotted "Inertia" (Sum of squared distances) against the number of clusters (K).
- The graph shows a sharp bend (an "elbow") at **K=5**. This indicates that 5 clusters capture the data patterns efficiently without making the model overly complex.

2. Cluster Interpretation Based on the visualization, we identified 5 distinct customer types:

- **Cluster 0 (Moderate/Average):** Average Income, Average Spending. (The "Standard" Customer).
- **Cluster 1 (High Income, Low Spending):** Earns a lot but spends little. (The "Saver" or "Miser").
- **Cluster 2 (Low Income, Low Spending):** Earns little and spends little. (The "Sensible" customer).
- **Cluster 3 (Low Income, High Spending):** Earns little but spends a lot. (The "Careless" spender).
- **Cluster 4 (High Income, High Spending):** Earns a lot and spends a lot. (The "Target" VIP customer).

3. Business Conclusion The marketing team should focus heavily on **Cluster 4 (VIPs)** with loyalty programs and premium offers. They should also monitor **Cluster 3** carefully as they are high spenders but financially vulnerable.